

PCMigration Reconciliation

v4.1.0 operational guide

Windows migration auditing, selective repair, verification, and rollback. Includes the experimental Windows 10 Start-menu recovery workflow.

Core capture schema 4.0: existing v4.0.0 captures remain compatible.

Start captures are separate: new StartMenu-1.0 or trusted legacy v2.5.4.

Execution host: 64-bit Windows PowerShell 5.1 as the intended migrated user.

Validate the package before use. Windows 10 tile rendering requires live verification.

Guide contents

1. Core workflow and commands
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Release date: September 30, 2026. Keep real captures, reports, and backups private. MIT license; no warranty.

PCMigration Reconciliation v4.1.0

PCMigration Reconciliation is a conservative Windows migration auditing and selective-repair toolkit. It captures independent evidence from a source and destination PC, explains what did not migrate, creates an inert repair plan, applies only individually approved and source-authorized actions, and preserves rollback evidence before mutation.

It is designed for experienced Windows users, administrators, technicians, and migration reviewers who want a transparent alternative to blindly copying profiles, registry hives, application databases, or credentials.

Release summary

Version 4.1.0 adds the Windows 10 Start-menu workflow omitted from v4.0.0: dual-mode XML capture, encoding-safe diagnostics, optional XML layout policy application, and the explicit tiles-only CloudStore fallback. The core reconciliation schema remains **4.0**: existing v4.0.0 core captures remain usable. Captures from v3.x remain incompatible. Start tiles use a separate capture with schema StartMenu-1.0; the tile restore also accepts trusted legacy v2.5.4 Start captures.

The added Start module is an experimental Windows 10 component. It does not restore Windows 11 pins. Read START-MENU.md before using it; it is deliberately separate from the automatic repair plan.

Most importantly, Windows shortcut files are no longer declared different merely because their opaque .lnk binary metadata differs. The comparison now uses normalized launch target, arguments, and working directory. A shortcut already present and functionally equivalent is listed in Shortcut-Reconciliation.csv, but is excluded from Repair-Plan.csv even if the raw files have different hashes.

Included files

File	Purpose
Test-PCMigrationPackage-v4.1.0.ps1	Parses scripts, verifies every packaged checksum, and runs regression tests.
Capture-PCMigrationState-v4.1.0.ps1	Captures applications, settings, registry state, shortcuts, Windows state, integration state, development tools, and manual/secure migration items.
Compare-PCMigrationState-v4.1.0.ps1	Validates both captures and generates reconciliation reports plus an inert repair plan. It never changes the destination.
Invoke-PCMigrationRepair-v4.1.0.ps1	Previews or applies individually approved, source-authorized repairs and creates backups before mutation.
Verify-PCMigrationState-v4.1.0.ps1	Recaptures the destination and repeats the comparison after repair.
Rollback-PCMigrationRepair-v4.1.0.ps1	Previews or restores file and narrow registry changes from a repair backup.
PCMigration.Common-v4.1.0.ps1	Shared PowerShell 5.1 and .NET implementation. Do not run directly.
RegistryBackup-v4.1.0.ps1	Optional native binary registry safety snapshots plus narrow targeted exports. Do not run directly.
PCMigration-Reconciliation-v4.1.0-Guide.pdf	Complete operational instructions and review guidance.
CAPABILITIES-AND-LIMITATIONS.md	Supported cases, special fixes, exclusions, and limitations.
Capture-StartMenu-v4.1.0.ps1	Separate Windows 10 XML and tile-grid capture.
Diagnose-Build-StartLayout-v4.1.0.ps1	Read-only tile diagnostics; explicit XML build/policy apply/remove/rollback.
Restore-StartTileGrid-v4.1.0.ps1	Preview/apply/rollback of one Windows 10 tile-grid subtree.
StartMenu.Common-v4.1.0.ps1	Start-specific validation, encoding, registry scope, and backup helpers.
START-MENU.md	Full Start workflow and compatibility requirements.
VALIDATION.md	What was checked and what still requires Windows validation.
tests/Test-StartMenuLogic.ps1	Registry-scope, XML encoding, and integrity regression tests.
.github/workflows/validate.yml	Windows PowerShell 5.1 validation on GitHub push/PR.
SHA256SUMS.txt	Unsigned checksum manifest; detects changes, not publisher identity.

Requirements

- Windows 10 or Windows 11.
- 64-bit Windows PowerShell 5.1. PowerShell 7 is not the supported execution host.
- The same migrated user should run the source and destination captures.
- Elevation is recommended for machine-wide coverage and required for administrative repairs and broad registry safety snapshots.
- Use a trusted, access-controlled drive. Captures contain detailed system and application metadata; optional registry snapshots are especially sensitive.
- Extract the release into its own clean folder. By default, unrelated files produce warnings; use `-StrictPackageContents` to reject them.

Safe operating model

1. Capture broadly enough to establish evidence.
2. Compare offline without changing either PC.
3. Install applications before restoring their settings.
4. Review shortcut reconciliation before approving any shortcut copy.
5. Approve individual actions only.
6. Preview with the same authorization switches that will be used for application.
7. Apply, sign out, verify, and retain the rollback folder.

Every generated `Repair-Plan.csv` begins with `Approved=N0`. Neither `-AllowHighRisk` nor `-AllowAdminChanges` approves a row; those switches only authorize already approved rows.

Complete workflow

Run all toolkit scripts from the extracted package directory in elevated 64-bit Windows PowerShell 5.1 unless a step explicitly says otherwise.

1. Validate the clean package

```
Set-ExecutionPolicy -Scope Process Bypass -Force
Get-ChildItem -LiteralPath $PWD -Filter '*.ps1' | Unblock-File
.\Test-PCMigrationPackage-v4.1.0.ps1
```

Expected result:

Package validation passed: PowerShell parser, SHA256 manifest, and v4.1.0 regression checks.

Unrelated files are warnings by default. To require an exact package-only directory:

```
.\Test-PCMigrationPackage-v4.1.0.ps1 -StrictPackageContents
```

Do not continue if a parser, missing-file, or hash failure is reported.

2. Capture the source PC

For Windows 10 pinned tiles, also run `Capture-StartMenu-v4.1.0.ps1` on the source into a separate new directory while the original layout is still present. Follow `START-MENU.md`. The regular core capture, including `-CaptureRegistrySafetyBackup`, is not a supported tile-restore input.

Close applications whose settings matter. The practical default is:

```
.\Capture-PCMigrationState-v4.1.0.ps1 `
  -OutputPath 'P:\PCMigration-Source-v4.1.0' `
  -InventoryMode Standard `
  -CaptureSettingsPayload `
  -SearchSecureFileCandidates `
  -StartStoppedInventoryServices
```

If portable applications exist outside the usual locations:

```
-AdditionalPortableRoots 'D:\Tools','C:\Utilities','C:\PortableApps'
```

Optional comprehensive registry safety backup:

```
.\Capture-PCMigrationState-v4.1.0.ps1 `
  -OutputPath 'P:\PCMigration-Source-v4.1.0' `
  -InventoryMode Standard `
  -CaptureSettingsPayload `
  -SearchSecureFileCandidates `
  -StartStoppedInventoryServices `
  -CaptureRegistrySafetyBackup
```

This adds native HKCU\Software and HKLM\SOFTWARE binary snapshots. They are forensic safety archives only, are never repair payloads, and must never be restored wholesale to another PC. Store the capture on encrypted, access-controlled media.

To obtain only the narrow targeted registry exports without the binary snapshots:

```
-CaptureRegistrySafetyBackup -RegistryBackupTargetedOnly
```

If a verified third-party application service owns a legitimate settings database, it can be stopped only during file capture and automatically restarted:

```
-QuiesceServiceName 'ExactVerifiedServiceName'
```

Never guess a service name. Security, Defender, Kaspersky, firewall, networking, credential, Search, AppX, VSS, installer, and core Windows services are denylisted.

3. Review the source capture

```
Import-Csv 'P:\PCMigration-Source-v4.1.0\Capture-Status.csv' |
Format-Table Collector,Status,Records,ElapsedSeconds,Message -AutoSize -Wrap
```

Investigate every Failed result. A Partial collector creates a comparison blind spot and is never interpreted as proof that an item is absent.

Review all captured shortcuts with useful relative locations:

```
Import-Csv 'P:\PCMigration-Source-v4.1.0\Applications\Shortcuts.csv' |
Sort-Object RelativeLocation |
Format-Table RelativeLocation,TargetPath,Arguments,TargetExists -AutoSize -Wrap
```

4. Capture the destination before repair

Run as the migrated user, preferably elevated:

```
.\Capture-PCMigrationState-v4.1.0.ps1 `
  -OutputPath 'C:\MigrationAudit\Destination-Before-v4.1.0' `
  -InventoryMode Standard `
  -SearchSecureFileCandidates `
  -StartStoppedInventoryServices
```

Do not use -CaptureSettingsPayload or -CaptureRegistrySafetyBackup on the destination.

5. Compare source and destination

```
.\Compare-PCMigrationState-v4.1.0.ps1 `
-SourceCapture 'P:\PCMigration-Source-v4.1.0' `
-DestinationCapture 'C:\MigrationAudit\Destination-Before-v4.1.0' `
-OutputPath 'C:\MigrationAudit\Reconciliation-Before-v4.1.0'
```

Open the HTML summary:

```
Start-Process 'C:\MigrationAudit\Reconciliation-Before-v4.1.0\Summary.html'
```

Review in this order:

1. Capture-Coverage.csv
2. Regression-Checks.csv
3. Shortcut-Reconciliation.csv
4. Shortcut-Gaps.csv
5. Likely-Migration-Gaps.csv
6. Missing-Applications.csv
7. Application-Settings-Gaps.csv
8. Windows-State-Gaps.csv
9. Manual-Secure-Actions.csv
10. Repair-Plan.csv

6. Confirm shortcut results before any repair approval

Show every shortcut and its logical destination result:

```
Import-Csv 'C:\MigrationAudit\Reconciliation-Before-v4.1.0\Shortcut-Reconciliation.csv' |
Sort-Object FunctionalStatus,RelativeLocation |
Format-Table RelativeLocation,DestinationPresent,FunctionalStatus,BinaryStatus,Risk,SourceTarget -AutoSize
-Wrap
```

Show only real functional gaps:

```
Import-Csv 'C:\MigrationAudit\Reconciliation-Before-v4.1.0\Shortcut-Gaps.csv' |
Sort-Object RelativeLocation |
Format-Table RelativeLocation,Status,DestinationPresent,TargetPath,Arguments -AutoSize -Wrap
```

Shortcut-Equivalent-Binary-Differences.csv is informational. Its rows were suppressed from the repair plan because the destination shortcut is already present and functionally equivalent. This is the expected location for shortcuts previously reported as high risk solely because the .lnk binary hashes differed.

7. Install missing applications first

Install required applications before restoring settings or shortcuts. Prefer compatible versions and vendor-supported installers. Exact source-correlated WinGet identities may appear as WinGetInstall actions, but IDs and publishers must still be reviewed.

Never copy live credential stores, browser session databases, Keeper sessions, Windows Hello/DPAPI material, antivirus databases, Store repositories, or unknown application databases.

8. Recapture and compare after application installation

```
.\Capture-PCMigrationState-v4.1.0.ps1 `
-OutputPath 'C:\MigrationAudit\Destination-PostApps-v4.1.0' `
-InventoryMode Standard `
-SearchSecureFileCandidates `
-StartStoppedInventoryServices

.\Compare-PCMigrationState-v4.1.0.ps1 `
-SourceCapture 'P:\PCMigration-Source-v4.1.0' `
-DestinationCapture 'C:\MigrationAudit\Destination-PostApps-v4.1.0' `
-OutputPath 'C:\MigrationAudit\Reconciliation-PostApps-v4.1.0'
```

Use the new post-application Shortcut-Reconciliation.csv and Repair-Plan.csv. Do not reuse or merge an earlier plan.

9. Approve selected actions

```
$PlanPath = 'C:\MigrationAudit\Reconciliation-PostApps-v4.1.0\Repair-Plan.csv'
$Plan = Import-Csv -LiteralPath $PlanPath

$Plan |
Sort-Object {[int]$_Priority},Risk,Method,Item |
Format-Table ActionId,Approved,Priority,Method,Risk,RequiresAdmin,Item -AutoSize -Wrap
```

After reviewing specific action IDs:

```
$ApprovedActionIds = @('A00012','A00018')

foreach($Row in $Plan){
    $Row.Approved = if($Row.ActionId -in $ApprovedActionIds){'YES'}else{'NO'}
}

$Plan | Export-Csv -LiteralPath $PlanPath -NoTypeInformation -Encoding UTF8
```

Edit only the Approved column. Do not change method, identity, artifact, destination, package, feature, capability, or hash fields.

10. Preview the approved repair

Low/medium current-user actions:

```
.\Invoke-PCMigrationRepair-v4.1.0.ps1 `
-SourceCapture 'P:\PCMigration-Source-v4.1.0' `
-PlanPath $PlanPath
```

If approved rows include high-risk or administrative actions, preview with the same gates required for application:

```
.\Invoke-PCMigrationRepair-v4.1.0.ps1 `
-SourceCapture 'P:\PCMigration-Source-v4.1.0' `
-PlanPath $PlanPath `
-AllowHighRisk `
-AllowAdminChanges
```

Preview mode makes no destination changes.

11. Apply the reviewed repair

```
.\Invoke-PCMigrationRepair-v4.1.0.ps1 `
-SourceCapture 'P:\PCMigration-Source-v4.1.0' `
-PlanPath $PlanPath `
-AllowHighRisk `
-AllowAdminChanges `
-Apply
```

The script records the exact backup folder on the destination desktop. Keep it until final verification and a stable operating period are complete.

12. Optional Windows 10 Start tiles; then sign out and sign back in

After apps and shortcuts are installed/reconciled, follow `START-MENU.md` to diagnose or restore tiles. Tiles are not listed in `Repair-Plan.csv`. The main verification checks apps/shortcuts/settings; verify tile rendering visually after sign-out/sign-in.

This reloads Explorer, Console Host, regional/clock, font, language, personalization, and shell state. Restart Windows when a feature, capability, installer, or vendor application requires it.

13. Verify

```
.\Verify-PCMigrationState-v4.1.0.ps1 `
-SourceCapture 'P:\PCMigration-Source-v4.1.0' `
-DestinationCapturePath 'C:\MigrationAudit\Destination-After-v4.1.0' `
-ReportPath 'C:\MigrationAudit\Reconciliation-After-v4.1.0'
```

Review `Summary.html`, `Regression-Checks.csv`, `Capture-Coverage.csv`, `Shortcut-Reconciliation.csv`, `Likely-Migration-Gaps.csv`, and `Manual-Secure-Actions.csv`.

14. Roll back if necessary

Preview:

```
.\Rollback-PCMigrationRepair-v4.1.0.ps1 `
-BackupPath 'C:\Users\USER\Desktop\PCMigration-v4.1.0-Backup-YYYYMMDD-HHMMSS'
```

Apply:

```
.\Rollback-PCMigrationRepair-v4.1.0.ps1 `
-BackupPath 'C:\Users\USER\Desktop\PCMigration-v4.1.0-Backup-YYYYMMDD-HHMMSS' `
-Apply
```

To remove registry subtrees that did not exist before repair, add `-RemoveNewRegistryKeys` only after reviewing the preview. Package installations, Windows features, and capabilities are recorded but are not automatically uninstalled.

Unicode and special-path behavior

Package text, CSV, and JSON outputs use Unicode-safe APIs and UTF-8; native registry exports use UTF-16. Start XML is decoded by the XML parser using its declared encoding, including UTF-8 without a BOM. Literal filesystem operations preserve the exact original path. Comparison normalization uses Unicode Form C only; compatibility characters are not collapsed. Therefore the legal Windows filename character μ (U+00B5 MICRO SIGN) survives capture, reporting, manifest hashing, payload copy, and repair authorization, while remaining distinct from μ (U+03BC GREEK SMALL LETTER MU). The validator contains a regression test for this behavior.

Reparse points are deliberately skipped, parent traversal is rejected, and repair destinations must resolve through an approved root token or shortcut root. The tool does not promise support for every legacy

application that mishandles Unicode internally.

What v4.0.0 omitted

- The separate Start-menu capture and dual XML export workflow from v2.5.4.
- Its XML diagnosis/build/policy application path, including the encoding-specific μ correction.
- The v2.5.6.1 tile-grid-only CloudStore fallback and its independent rollback path.

Version 4.1.0 consolidates those Start capabilities with revised scope checks, preview gates, and policy backups. It does not transplant `systempartitionindex` or the whole CloudStore hive. Full firewall policy import, taskbar-pin transplantation, and generic illegal-path repair are still outside this toolkit. Its Unicode handling preserves valid Unicode names; it does not make genuinely invalid Windows paths valid.

Important limitations

- This is reconciliation and selective repair, not disk imaging, profile cloning, or a guarantee of identical PCs.
- Credentials, private keys, browser tokens, DPAPI state, Windows Hello, passkeys, EFS keys, protected Store data, and antivirus runtime databases are never generically migrated.
- Drivers, services, tasks, VPNs, printers, shares, firewall rules, ODBC DSNs, certificates with private keys, and licenses are primarily inventoried for deliberate/manual handling.
- Captures can contain personal and security-sensitive metadata. Optional binary registry snapshots are highly sensitive.
- A hash difference proves bytes differ; it does not prove the source should overwrite the destination.
- Applications should be installed before their settings are restored.
- Windows build, architecture, application-version, policy, domain, and hardware differences can make source settings inappropriate.
- Always maintain independent backups. Use at your own risk; no warranty is provided.

See `CAPABILITIES-AND-LIMITATIONS.md` and the PDF guide for the detailed support matrix and safety model.

License

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Windows 10 Start-menu capture and restoration

This module supplies the Start workflow omitted from v4.0.0. It is separate from the main reconciliation capture and repair plan. Run as the intended user in **64-bit Windows PowerShell 5.1**, in that user's interactive session. Windows 11 pins and Windows Server are excluded.

Choose the right operation

Need	Command	Capture/input
Audit or restore Start-menu shortcuts	Main capture/compare/repair workflow	Core schema 4.0 capture
Preserve Windows 10 pins for future migration	Capture-StartMenu-v4.1.0.ps1	Run on source; separate new folder
Inspect link/ID exports and unresolved tiles	Diagnose-Build-StartLayout-v4.1.0.ps1	Separate Start capture
Apply the exported XML layout	Diagnosis command with -ApplyPolicy -Apply	XML export; Windows edition/policy support required
Restore only the captured tile grid when XML omitted pins	Restore-StartTileGrid-v4.1.0.ps1 -Apply	Captured tile-grid registry export

Start-menu shortcuts and pinned tiles are different state. Install applications and reconcile shortcuts first. A regular core capture, including its optional registry safety snapshots, is **not** an input for the Start restore command.

1. Capture the original source layout

Run this on the source while its desired layout is still present:

```
.\Capture-StartMenu-v4.1.0.ps1 `
-OutputPath 'P:\PCMigration-StartMenu-Source-v4.1.0'
```

Use a new, nonexistent directory. The capture contains:

- Meta.json and Capture-Status.csv.
- StartLayout-Links.xml and StartLayout-IDs.xml, if each export succeeds.
- Get-StartApps.csv and StartMenu-Shortcuts.csv with relative locations.
- Captured .lnk and .url files under StartMenu\User and StartMenu\Common.
- CloudStore\CloudStore-KeyIndex.csv and successfully exported Key-###.reg files for curated tile-grid keys only.
- SHA256SUMS.txt for all captured artifacts. Do not edit the capture.

Review Capture-Status.csv. XML export and CloudStore export are independent: an XML failure does not necessarily prevent tile-grid recovery. An XML success does not establish that all visible pins were exported. Compare both exports with the source layout. Exporting live state is not an atomic Windows snapshot.

The new module uses capture schema StartMenu-1.0. Trusted earlier Capture-StartMenu-v2.5.4.ps1 captures remain accepted. Legacy captures lack a manifest and therefore produce a warning. Earlier reconciliation captures are not interchangeable with Start captures.

2. Prepare the destination

Install missing applications, run the main comparison, review Shortcut-Reconciliation.csv, and apply only the necessary approved shortcut actions. A functionally equivalent shortcut already on the destination needs no copy.

If you may apply an XML policy, preserve the destination's original Start layout first:

```
.\Capture-StartMenu-v4.1.0.ps1 `
  -OutputPath 'C:\MigrationAudit\PCMigration-StartMenu-Destination-Before-v4.1.0'
```

The XML policy backup preserves the three policy values only; it does not preserve the previous tile grid. The binary fallback automatically creates its own grid-and-policy backup.

3. Diagnose before restoring

```
.\Diagnose-Build-StartLayout-v4.1.0.ps1 `
  -CapturePath 'P:\PCMigration-StartMenu-Source-v4.1.0' `
  -OutputPath 'C:\MigrationAudit\StartReports-v4.1.0'
```

Use a new output directory. This writes reports; it does not change Start, copy shortcuts, or stop Explorer.

Open StartTiles-Links-Diagnostics.csv and StartTiles-IDs-Diagnostics.csv. Each row contains the group, tile type, relative shortcut location, destination link, link presence, application IDs, and identity resolution.

To investigate a particular missing app, add -NameMatchPattern 'Blockstream|Green'. This is a general regex filter, not a hardcoded application dependency. If a tile is absent from both XML exports despite being visible on the source, XML editing cannot recover its omitted pin.

Legacy captures do not record source profile roots. If they contain absolute old-user paths, review and resolve those through the main shortcut workflow. New captures record the source roots for destination-user path remapping.

4A. XML layout route

Build corrected XML and diagnose its destination identities:

```
.\Diagnose-Build-StartLayout-v4.1.0.ps1 `
  -CapturePath 'P:\PCMigration-StartMenu-Source-v4.1.0' `
  -OutputPath 'C:\MigrationAudit\StartReports-XML-v4.1.0' `
  -BuildCorrectedLayout
```

This writes StartLayout-Corrected.xml. It remaps recorded source-user roots and updates links only when the destination file exists. It does not invent missing pins or stage/copy source shortcuts. Taskbar layout elements are omitted from this Start-only output. Use -LayoutMode IDs if the ID export better represents the desired source layout.

Preview policy installation (new report directory):

```
.\Diagnose-Build-StartLayout-v4.1.0.ps1 `
  -CapturePath 'P:\PCMigration-StartMenu-Source-v4.1.0' `
  -OutputPath 'C:\MigrationAudit\StartReports-PolicyPreview-v4.1.0' `
  -ApplyPolicy
```

Apply after review (another new report directory):

```
.\Diagnose-Build-StartLayout-v4.1.0.ps1 `
  -CapturePath 'P:\PCMigration-StartMenu-Source-v4.1.0' `
  -OutputPath 'C:\MigrationAudit\StartReports-PolicyApply-v4.1.0' `
  -ApplyPolicy `
  -Apply
```

This stores the XML under the user's LocalAppData and sets three current-user Explorer policy values: LockedStartLayout, StartLayoutFile, and ReapplyStartLayoutEveryLogon. It backs up the prior values, including absence and value type, on the user's actual Desktop folder.

Sign out completely and sign back in. A full layout policy locks customization while enabled. After checking the applied layout, preview clearing it:

```
.\Diagnose-Build-StartLayout-v4.1.0.ps1 -RemovePolicy
```

Then clear it and sign out/in again:

```
.\Diagnose-Build-StartLayout-v4.1.0.ps1 -RemovePolicy -Apply
```

Removal backs up the current policy first. It clears only those three values and does not restore the prior tile layout. Organization-managed policy may be reapplied; the module does not bypass it. XML policy support varies with Windows edition and configuration.

Policy rollback, using the exact printed policy backup:

```
.\Diagnose-Build-StartLayout-v4.1.0.ps1 `
  -RollbackBackup 'C:\Users\USER\Desktop\PCMigration-StartPolicy-Backup-EXACT-FOLDER'

.\Diagnose-Build-StartLayout-v4.1.0.ps1 `
  -RollbackBackup 'C:\Users\USER\Desktop\PCMigration-StartPolicy-Backup-EXACT-FOLDER' `
  -Apply
```

Policy rollback restores policy values only. Keep the destination Start capture from step 2 if you may need its former tile grid. Policy XML files and report files remain as audit artifacts; rollback does not delete them.

4B. Tiles-only binary fallback

Use this route when XML export omitted actual source pins and you have a trusted tile-grid capture. This is an **unsupported, experimental Windows 10 binary-state operation**. It is not automatically scheduled in Repair-Plan.csv.

Preview:

```
.\Restore-StartTileGrid-v4.1.0.ps1 `
  -CapturePath 'P:\PCMigration-StartMenu-Source-v4.1.0'
```

Apply:

```
.\Restore-StartTileGrid-v4.1.0.ps1 `
  -CapturePath 'P:\PCMigration-StartMenu-Source-v4.1.0' `
  -Apply
```

For your existing legacy capture, substitute its real folder, for example H:\PCMigration-StartMenu-v2.5.4. You do not need to rerun a source capture if that trusted folder already contains Meta.json, the CloudStore index, and its successful tile-grid .reg export.

The command:

1. Verifies capture hashes when supplied and validates every registry header against exactly one curated tile-grid subtree.

2. Locates one destination tile-grid key and retains its dynamic identity/GUID, even if the source and destination identities already match.
3. Remaps registry header paths only. Binary/string value data is not rewritten.
4. Backs up the destination tile grid and the three Start policy values before mutation.
5. Stops shell processes in the current session, clears those policy values, replaces the selected grid, and immediately compares registry exports.
6. Attempts automatic grid-and-policy rollback if import or verification fails; restarts Explorer and removes temporary .reg staging in finally.

It does not copy shortcuts, restore taskbar pins, transplant systempartitionindex, replace unrelated CloudStore state, or install apps. Temporary .reg staging is required for the native registry import and lives under %TEMP%, outside the capture/package.

Different Windows build numbers are blocked by default. -ForceDifferentBuild is an explicit unsupported Windows 10 override; it never enables Windows 11. If no destination grid exists, pin one tile and sign out/in first. If multiple grids are listed, review them and supply -DestinationTileGridChildName rather than guessing.

5. Verify visually after sign-out/in

Check group names, tile order/size, missing pins, and application launches. A successful immediate registry comparison confirms imported state, not correct shell rendering or application identity resolution. The main verification command does not certify the tile layout.

Keep the exact backup folder printed by the tile script. It is independent from the core repair backup.

Preview tile rollback:

```
.\Restore-StartTileGrid-v4.1.0.ps1 `
  -RollbackBackup 'C:\Users\USER\Desktop\PCMigration-StartTileGrid-Backup-EXACT-FOLDER'
```

Apply rollback and sign out/in:

```
.\Restore-StartTileGrid-v4.1.0.ps1 `
  -RollbackBackup 'C:\Users\USER\Desktop\PCMigration-StartTileGrid-Backup-EXACT-FOLDER' `
  -Apply
```

Rollback requires the same user, computer, and Windows build and validates the backup manifest. New rollback does not consume older v2.5.6.1 backup folders; keep that older script with any existing old-format backups.

Unicode fix and references

Start XML is read with an XML reader that honors its encoding declaration/BOM and prohibits DTDs. UTF-8 without a BOM therefore preserves μ (U+00B5) rather than producing Âμ. This is distinct from the core toolkit's exact Unicode filename handling. μ and μ (U+03BC) remain different characters.

Microsoft documents XML export/deployment and policy behavior; it does not document or support this binary CloudStore migration fallback:

- [Customize the Start layout](#)
- [Start policy settings](#)

For release validation status, see VALIDATION.md. Protect all captures/backups and never upload actual user state to GitHub.

What was omitted from v4.0.0

This audit compares the actual consolidated v4.0.0 distribution with the separate v2.5.4 StartMenu, v2.5.6.1 StartCloudStore, and v2.5.2 WindowsState packages. It distinguishes restored Start capabilities from older standalone operations that remain outside the conservative core repair engine.

Earlier capability	Status in v4.0.0	Status in v4.1.0
Start-menu shortcut inventory/copy	Included with semantic reconciliation and readable locations	Retained; approved copies use the core repair plan
Windows 10 Start XML capture in link and application-ID modes	Separate capture omitted	Included in Capture-StartMenu
XML encoding fix for µTorrent becoming ÅµTorrent	Core Unicode-path tests existed; the distinct XML decoder fix was omitted with the Start module	Included with encoding-aware XML loading and a BOM-less UTF-8 regression
Start tile diagnostics and corrected XML build	Separate script omitted	Included in Diagnose-Build-StartLayout; no automatic shortcut-tree copying
Explicit XML layout policy application/removal	Separate script omitted	Included with preview/apply gates and selected policy-value backup/rollback
Exact tile-grid fallback when XML omitted a source pin	v2.5.6.1 fallback omitted	Included in Restore-StartTileGrid with subtree/header validation and independent rollback
Full firewall .wfw import	Firewall rules inventoried/reported; the separate full-policy importer was not included	Still manual/outside the core repair engine; no full-policy replacement added
Dedicated taskbar-pin XML reconstruction	UserPinned shortcuts inventoried, but no dedicated pin-layout applicator	Still outside the added Start-only module; copying a shortcut does not establish a taskbar pin
Broad Explorer Streams/ShellBags/MRU imports from the older WindowsState script	Not carried forward as generic repair; Explorer evidence and selected settings remain	Still outside automatic repair; broad user history is not transplanted

The complete v4.1.0 package is complete for its documented scope, not a union of every experimental command in every older archive. Unreviewed whole-tree shortcut copies and broader legacy imports were not included merely to make the archive appear comprehensive.

The optimized registry safety backup, core capture/compare/repair/verify/rollback, shortcut equivalence, protected-state exclusions, service restoration, WSL/certificate/WLAN fallbacks, HTML hotfix, and exact Unicode filename handling remain consolidated.

Use START-MENU.md for the restored Start workflow and CAPABILITIES-AND-LIMITATIONS.md for the current support boundary.

Capabilities, special fixes, and limitations

Designed for

- Windows 10 and Windows 11 migration reconciliation.
- One source PC and one destination PC, run in the intended migrated user's interactive session.
- 64-bit Windows PowerShell 5.1 with inbox Windows/.NET components.
- Technicians and advanced users who will inspect evidence and approve individual changes.
- Offline comparison and default-deny destination repair.

Broad cases covered

- Applications: current/all-user AppX/MSIX evidence, WinGet export/list, 32/64-bit uninstall registrations, Start identities, shortcuts, and portable executable footprints.
- Application settings: recursively inventoried configuration/script files under AppData, LocalAppData, and optionally ProgramData; content-screened source payloads; scoped registry families; browser profile and extension inventories.
- User and shell state: Console Host, regional/language/time zone, 24-hour clock, Explorer/personalization evidence, fonts, default app associations, PowerShell profiles/modules/policies, Windows Terminal, and K-Lite/MPC-HC/Icaros/LAV-related state.
- Integration state: services, scheduled tasks, environment/startup values, printers, mapped drives, VPN/Wi-Fi metadata, optional features, capabilities, firewall rules, SMB shares, ODBC DSNs, certificates, and selected system files.
- Development state: installed tools, Git configuration evidence, WSL distribution inventory with legacy syntax fallback, and VS Code extensions.
- Secure/manual state: certificate/private-key warnings, authentication vaults, browser sessions, licenses, EFS/BitLocker/SSH/GPG/VPN/code-signing keys, and user-selected secure-file candidates.

Special fixes consolidated in v4.1.0

1. Shortcut semantic reconciliation: .lnk files are compared by launch target, arguments, and working directory. Binary-only changes no longer create repair actions.
2. Human-readable shortcut locations: capture and comparison outputs include paths such as Start Menu (all users)\Programs\Vendor\Tool.lnk.
3. Complete shortcut ledger: Shortcut-Reconciliation.csv shows equivalent, missing, different, and destination-only shortcuts; Shortcut-Gaps.csv contains only functional gaps.
4. Unicode-safe paths: U+00B5 survives capture/copy/reporting and is kept distinct from U+03BC; a regression test enforces this.
5. PowerShell 5.1 HTML compatibility: report fragments are explicitly joined before ConvertTo-Html -Body.
6. Native registry safety snapshots: optional sequential reg.exe save snapshots replace slow broad textual backup. They are forensic-only and cannot be authorized as automatic repair payloads.
7. Value-level registry comparison: selective comparison uses deterministic value digests rather than payload-file hashes.
8. Protected registry policy: Wi-Fi/WWAN/EAP authentication, compatibility telemetry, AppModel, credential, and other protected branches are explicit exclusions instead of misleading failures.

9. Volatile/protected file policy: Defender, Kaspersky, Search indexes, notifications/activity stores, diagnostics databases, and thumbnail/icon caches are excluded before enumeration or hashing.
10. Shared-read hashing: ordinary live files are opened with read/write/delete sharing and bounded retries.
11. Controlled service quiescing: verified third-party application services can be stopped around settings capture and are restored in finally; critical/security services are denylisted.
12. WLAN inventory recovery: WLAN AutoConfig can be started temporarily and returned to its original state.
13. Certificate fallback: read-only .NET X509Store enumeration continues when the PowerShell provider fails.
14. WSL compatibility: modern verbose listing falls back to legacy `wsl.exe --list` output with null-character normalization.
15. Resilient package validation: unlisted files warn by default and fail only with `-StrictPackageContents`; packaged file hashes and parser failures always remain fatal.
16. Source authorization: repair methods, destinations, identities, artifacts, and hashes are reconstructed from the source capture, not trusted solely from the editable plan.
17. Resumable recovery evidence: backup manifests and repair results are persisted during application, and existing narrow registry subtrees are restored exactly after cleanup.

Explicitly not handled automatically

- Whole-profile or whole-registry restoration.
- Automatic restoration of HKCU\Software or HKLM\SOFTWARE binary snapshots.
- Passwords, cookies, tokens, credentials, private keys, DPAPI, Windows Hello, passkeys, biometric enrollment, or authenticator secrets.
- Arbitrary application databases or version-bound caches.
- Antivirus/security databases or disabling security products to access them.
- Generic driver installation, service/task recreation, firewall/share changes, certificate private-key import, printer-driver deployment, or license transfer.
- Automatic uninstallation of packages/features/capabilities during rollback.
- Resolving business-policy questions about which destination-specific settings should win.

Operational limitations

- Core capture schema 4.0 is retained: v4.0.0 core captures remain compatible. Pre-v4 core captures are incompatible.
- Destination inventory depth must be at least as deep as source inventory depth.
- Incomplete collectors create Unknown coverage; they do not prove absence.
- Reparse points are skipped to avoid loops and unintended volumes.
- Machine and application version differences can make technically copyable settings undesirable.
- Store and packaged applications often require reinstall/sign-in/vendor sync rather than file copying.
- Some application installers regenerate shortcuts with equivalent behavior but different icons/descriptions; these are intentionally treated as functionally equivalent.
- The package has no telemetry and no remote orchestration; operators must securely transfer captures and protect report contents.

Security and privacy

Treat every capture, report, repair plan, backup directory, and especially every .hiv snapshot as sensitive. Use BitLocker or equivalently protected storage, restrict ACLs, do not commit captures to Git, and securely retire them after verification. The included .gitignore excludes typical generated folders, but it is not a substitute for reviewing staged content.

Windows 10 Start module added in v4.1.0

- Separate capture of both XML export modes, Get-StartApps identities, .lnk/.url shortcut trees, and narrowly selected curated tile-grid exports.
- XML decoding respects the file encoding: BOM-less UTF-8 preserves µ instead of producing Åµ.
- Readable relative shortcut locations, remapping of recorded source-user roots, and diagnostics for links versus application IDs.
- XML build changes only the output document. It does not copy shortcuts; reviewed shortcut repair remains in the core toolkit.
- XML application is preview-only without -Apply, uses three current-user policy values, and keeps a policy backup. It locks the layout until policy is removed. Policy rollback is not tile-grid rollback.
- Explicit binary tile-grid fallback retains the destination key identity and validates every registry header. It changes only one curated Start tile-grid subtree and three current-user Start policy values.
- Existing source/destination key identities may match. Header mapping does not rewrite value data.
- Shell stopping is confined to the current session; apply failures attempt to restore the grid and policy; Explorer restart and temporary-file cleanup run in finally.
- New captures/backups have checksums; a trusted legacy v2.5.4 Start capture remains accepted with a warning. Checksums are not digital signatures.
- The module is experimental, for Windows 10 clients only, with matching build numbers by default. Windows 11 pins, Start taskbar pins, CloudStore systempartitionindex, and unrelated CloudStore settings are excluded.
- Immediate registry verification cannot prove that every tile will render or launch. Install applications first and verify after sign-out/in.
- Managed policy or Windows edition restrictions may prevent XML application. The toolkit does not bypass organization policy.

Release validation and test boundaries

Checks performed while preparing v4.1.0

- Reviewed the actual v4.0.0 suite, v2.5.4 Start capture/diagnosis, and v2.5.6.1 tile fallback.
- Kept the core schema at 4.0; updated script dependency names and tool version metadata.
- Checked script structure with a third-party PowerShell grammar, accounting for grammar limitations in unchanged inherited code. This is not the Microsoft PowerShell parser.
- Checked UTF-8 BOM and CRLF encoding for release PowerShell sources, relative dependencies, safe ZIP paths, exact manifest coverage, archive extraction, and SHA-256 values.
- Created and rendered the updated PDF guide and visually inspected its pages.

Required Windows checks

The authoring workspace is Linux and has no Microsoft PowerShell runtime. Windows PowerShell 5.1 parser/regression execution, shell rendering, registry imports, policy application, and real source/destination migration were not performed here. Do not describe this release as Windows integration-tested.

On Windows, run `Test-PCMigrationPackage-v4.1.0.ps1` before use. It verifies hashes and parses sources before executing regression code. It includes Start logic checks for authorized registry scope, header-only mapping, same-identity mapping, BOM-less UTF-8 XML, DTD rejection, and changed capture files, alongside the inherited core regression checks. The tests do not change Start, policy, or the registry.

The included GitHub workflow executes the package validator with **Windows PowerShell 5.1** on a Windows runner. It stages the exact tracked repository source without `.git`, then validates with `-StrictPackageContents`. The runner validates parser/logic behavior, not a live Windows 10 tile restore.

Before promoting the experimental Start module to a stable public release, validate capture, preview, apply, sign-out/in rendering, failure recovery, and rollback using disposable Windows 10 user profiles on matching builds. Keep the experimental designation until that evidence exists.

Editing repository source

Any source or documentation edit changes the packaged checksums. Update `SHA256SUMS.txt` deliberately when preparing the next release; do not rewrite it merely to suppress a validation failure. Run strict validation against the release staging directory. New captures/backups have independent manifests and must never be edited.

Checksums detect changes relative to their manifest. They are not signatures and do not authenticate the maintainer.

GitHub publishing text

Repository name

PCMigration-Reconciliation

Short description

Windows 10/11 migration auditing and selective repair, with an experimental Windows 10 Start-tile recovery module. PowerShell 5.1 and inbox .NET APIs.

Suggested topics

windows powershell migration windows-11 windows-10 system-administration backup registry winget appx audit disaster-recovery

Project summary

PCMigration Reconciliation compares fresh source and destination captures across applications, settings, shortcuts, registry state, Windows/user configuration, integration state, development tools, and secure/manual migration items. It generates transparent CSV/HTML evidence and an inert repair plan. Repairs remain preview-only until individually approved and are independently authorized against the source capture before application. Destination files and narrow registry subtrees are backed up before mutation, followed by fresh verification and optional rollback.

The project intentionally avoids blind profile copies, whole-hive restoration, credential transplantation, protected security databases, and generic copying of unknown application databases.

v4.1.0 release summary

- Adds the Windows 10 Start workflow omitted from v4.0.0: dual-mode XML capture, encoding-safe diagnosis/build, optional XML policy application, and tiles-only CloudStore restore/rollback.
- Preserves the destination tile-grid identity, validates all registry headers, remaps headers only, accepts matching identities, and backs up the selected policy values.
- New Start captures/backups include integrity manifests. Trusted legacy v2.5.4 Start captures remain accepted with a warning.
- Fixes BOM-less UTF-8 XML decoding so μ does not become $\hat{\mu}$; keeps the core μ/μ distinction.
- Shortcut copying stays in the individually reviewed core repair plan. Start restore does not blindly copy trees.
- Adds Windows PowerShell 5.1 GitHub validation and logic regressions for scope, mapping, encoding, and capture tampering.
- Retains core schema 4.0; existing v4.0.0 source/destination captures remain usable.
- Updates the full guide, compatibility documentation, omission audit, security notes, and validation boundaries.

Release designation and limitations

Publish the initial v4.1.0 release as a **prerelease** until the Windows 10 Start module has been exercised on disposable profiles. Package structure/hash checks and static source checks are complete; Windows PowerShell and live Start application were not run in the Linux authoring workspace. The included GitHub validation tests parser/logic behavior, not tile rendering. See `VALIDATION.md`.

The core toolkit supports Windows 10/11. The Start module is Windows 10 only; binary CloudStore recovery is unsupported by Microsoft, with matching builds required by default. Existing v4.0.0 core captures remain compatible; pre-v4 core schemas remain incompatible. Start inputs are separate captures, not core captures or broad registry snapshots. Full firewall import, dedicated taskbar-pin restore, broad Explorer history/ShellBags restoration, and credential transplantation are not new automatic repairs.

Do not claim that `SHA256SUMS.txt` is digitally signed. Protect real captures/reports/backups and never publish them.

Files to publish

- All versioned `.ps1` source files.
- `README.md`, `START-MENU.md`, `OMISSION-AUDIT.md`, `VALIDATION.md`, `CAPABILITIES-AND-LIMITATIONS.md`, `CHANGELOG.md`, `SECURITY.md`, `LICENSE`, and `.gitignore`.
- `tests/` and `.github/workflows/validate.yml`.
- `docs/PCMigration-Reconciliation-v4.1.0-Guide.pdf`.
- `SHA256SUMS.txt` and the release ZIP SHA-256 value.

Do not publish source/destination captures, comparison reports, edited repair plans, repair backups, registry snapshots, or real diagnostic output.

Practical upload sequence

1. Extract the ZIP. Use the **contents** of `PCMigration-Reconciliation-v4.1.0` as the repository root, so `README` and scripts are at the top level.
2. Include `.gitignore` and `.github/workflows/validate.yml`; browser uploads may hide dot folders, so verify they are present.
3. Commit only `source/docs/tests/manifest`. Use the release ZIP as a Release asset rather than replacing the source tree with a ZIP file.
4. Allow the Windows validator workflow to finish and resolve failures before promoting the release.
5. Create tag `v4.1.0`, paste the release summary above, and attach the ZIP, standalone PDF, and external release checksum file. Keep the prerelease designation until live Windows 10 validation exists.

The package was prepared for upload; no repository was created or published during authoring.